MODULE 1: Open Source Software

Open source software is computer software whose source code is freely distributed with a license to study, change, and further distribute it to anyone for any purpose. This allows for greater collaboration, flexibility, and innovation in the software development process.

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What is Open Source?

1. **Free Distribution**

   Open source software can be freely copied, distributed, and used by anyone for any purpose.

2. **Accessible Source Code**

   The source code of open source software is publicly available, allowing users to study, modify, and improve upon it.

3. **Collaborative Development**

   Open source software is typically developed through a collaborative process, with contributions from a community of developers.

4. **Flexible and Customizable**

   Users can adapt open source software to their specific needs and requirements, as they have access to the source code.

Examples of Open Source Software

Operating Systems

Linux, OpenSolaris, FreeBSD, Minix

Web Browsers

Firefox, Chrome, Opera, Safari

Productivity Tools

LibreOffice, OpenOffice, GIMP, Inkscape

Rights of Open Source Software

1. ### Free Redistribution

   The license shall not restrict any party from selling or giving away the software as a component of an aggregate software distribution containing programs from several different sources.

2. ### Access to Source Code

   The program must include source code, and must allow distribution in source code as well as compiled form.

3. ### Derived Works

   The license must allow modifications and derived works, and must allow them to be distributed under the same terms as the license of the original software.

4. ### Integrity of Author's Source Code

   The license may restrict source-code from being distributed in modified form only if the license allows the distribution of "patch files" with the source code for the purpose of modifying the program at build time.

Promises of Open

Better Quality

Open source software often benefits from a large community of developers who can identify and fix issues more quickly.

Higher Reliability

The collaborative nature of open source development and the ability to access the source code can lead to more reliable and secure software.

More Flexibility

Users can customize open source software to meet their specific needs and requirements, as they have access to the source code.

Lower Cost

Open source software is typically available for free or at a much lower cost compared to proprietary software.

Criteria for Open Source Software

1

Free Redistribution

The license shall not restrict any party from selling or giving away the software as a component of an aggregate software distribution containing programs from several different sources.

2

Access to Source Code

The program must include source code, and must allow distribution in source code as well as compiled form.

3

Derived Works

The license must allow modifications and derived works, and must allow them to be distributed under the same terms as the license of the original software.

Open Source Principles

Free Redistribution

The license shall not restrict any party from selling or giving away the software.

Access to Source Code

The program must include source code and allow distribution in source code form.

Derived Works

The license must allow modifications and derived works to be distributed.

Integrity of Author's Source Code

The license may restrict source-code from being distributed in modified form only if the license allows the distribution of "patch files".

Open Standards for Software

1 **Interoperability**

Open standards must not prohibit conforming implementations in open source software.

2 **Collaborative Development**

Open standards are typically developed through a collaborative process, with contributions from a community of stakeholders.

3 **Community Adoption**

Open standards are widely adopted and used by a community of developers and users, ensuring their longevity and relevance.

Free and Open Source Software (FOSS)

Definition

FOSS refers to computer software that is free and open source, allowing users to freely use, copy, and modify the software based on their needs or requirements.

Examples

Some of the best-known examples of FOSS include the Linux kernel, the BSD and Linux operating systems, the GNU Compiler Collection, the MySQL database, the Apache web server, and the Sendmail mail transport agent.

Benefits

FOSS offers benefits such as better quality, higher reliability, more flexibility, and lower cost compared to proprietary software.

The FOSS

1

Collaboration

The FOSS community is built on a collaborative model, where developers from around the world contribute to the development and improvement of the software.

2

Shared Knowledge

FOSS projects encourage the sharing of knowledge and expertise, allowing the community to collectively solve problems and enhance the software.

3

Innovation

The open and collaborative nature of FOSS fosters innovation, as developers can build upon existing work and explore new ideas without the constraints of proprietary software.